

Bailicai: A Domain-Optimized Retrieval-Augmented Generation Framework for Medical Applications

Long Cui, Yongbin Liu*, Chunping Ouyang, Ying Yu, Jiangtao Zhang, Yaping Wan, and Fei Yang

Abstract: Large language models (LLMs) excel in various natural language processing tasks and are increasingly applied in specialized fields like medicine. However, their deployment in the medical domain is challenged by limited domain-specific data and the tendency to generate inaccurate information, known as “hallucinations.” While domainspecific fine-tuning has improved open-source LLMs, they still underperform compared to proprietary models like ChatGPT and PaLM. To address this gap, retrieval-augmented generation (RAG) techniques have been explored to enhance LLMs by integrating external knowledge bases. Nevertheless, the success of RAG depends on the quality of retrieved documents, and its application within the medical field remains in the early stages. In this paper, we introduce the “Bailicai” framework as an exploratory approach to integrating RAG with LLMs in the medical field. The framework employs fine-tuning to improve the RAG process, where “falsely relevant” and “completely irrelevant” interference documents are intentionally included in the training data. This enables Bailicai to develop the ability to assess the quality of retrieved documents and selectively incorporate them. The framework is organized into four modules: (1) medical knowledge injection, (2) self-knowledge boundary identification, (3) directed acyclic graph task decomposition, and (4) retrieval-augmented generation. Through the synergy of these modules, Bailicai achieves superior performance on multiple medical benchmarks, outperforming existing large models in the medical domain, RAG-based methods, and proprietary models such as GPT-3.5. Furthermore, Bailicai effectively mitigates the hallucination problem common in LLMs applied to medical tasks and enhances the robustness of RAG when dealing with irrelevant or misleading documents, enabling more accurate information retrieval and integration.

Key words: large language models (LLMs); retrieval-augmented generation (RAG); domain-specific language models

1 Introduction

Large language models (LLMs), trained on extensive

datasets, demonstrate the ability to generate plausible responses through their internal parameterized knowledge and few-shot capability^[1, 2], achieving

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performance comparable to that of human experts in complex natural language processing tasks and specialized domains^[3–6]. Recent studies, including those on GPT-4^[3, 7] and Med-PaLM 2^[8], indicate that these proprietary LLMs can achieve expert-level performance on specific medical benchmarks. The sensitive nature of medical data, however, raises concerns regarding patient privacy risks associated with these closed-source models^[9], thereby prompting heightened interest in the application of open-source LLMs in the medical domain. Nevertheless, open-source models continue to exhibit significant performance disparities compared to their closed-source counterparts in medical applications. These open-source models enhance their performance through pre-training and fine-tuning on medical datasets^[10–13], with their performance ceiling dependent on the quality of the datasets, employed strategies, underlying models, and computational resources^[4]. Despite pre-training and fine-tuning, the parameterized knowledge within the models remains insufficient to comprehensively address all medical issues, and the computational cost of retraining continues to be substantial^[14]. When presented with novel medical queries, these models may generate responses that deviate from established medical knowledge, a phenomenon commonly referred to as “Hallucination”^[15, 16].

To address these challenges, retrieval-augmented generation (RAG) technology enhances the knowledge base of models by retrieving information from external sources and incorporating it into the prompts of generative language models in a non-parametric manner^[17, 18]. Nevertheless, the integration of RAG techniques with LLMs presents several challenges, including the optimization of retrieval processes and their timing, effective utilization of contextual information, and mitigation of noise in retrieved documents^[19, 20]. Research efforts have been initiated to address these challenges, encompassing adaptive retrieval methods (e.g., FLARE^[21]), iterative retrieval (e.g., ITER-RETGEN^[22]), and fine-tuning adaptations for RAG tasks (e.g., Self-RAG^[23] and RA-ISF^[24]). In addition, the integration of RAG and LLM in the medical field is still in its infancy^[25], and leveraging the strengths of both to bridge the performance gap between open-source and closed-source models is a research direction that has not yet been fully explored.

Current research predominantly focuses on evaluating the impact of various components in RAG systems—including the LLMs, corpora, and retrieval models—through systematic assessment of their specific effects on performance^[26].

In response to these challenges, we propose the “Bailicai” framework, which efficiently integrates medical tasks with RAG technology. By integrating RAG with open-source large language models, Bailicai achieves a synergistic effect^[27] that substantially improves performance in medical applications. The core architecture of the Bailicai framework, illustrated in Fig. 1, consists of four primary components: (1) medical knowledge injection, which enhances the model by integrating domain-specific insights; (2) self-knowledge boundary identification, enabling the model to discern when to rely on its internal knowledge; (3) directed acyclic graph task decomposition, a novel approach for systematically decomposing complex queries into manageable sub-tasks; and (4) retrieval-augmented generation, which improves response accuracy by incorporating pertinent external information. During the data construction stage of Bailicai, we employ Self-Instruct and the model-oriented directive data filtering (MoDS^[28]) methods to further enhance and filter the original UltraMedical dataset, utilizing the Meta-Llama-3-70B model for various tasks. Consequently, fine-tuning datasets tailor to three distinct tasks were constructed. To enhance the robustness of the RAG method against interference, we introduce the “hard negative mining” strategy, which incorporates “falsely relevant” and “completely irrelevant” retrieval documents into the dataset as contextual knowledge grounded in contextual reasoning^[29, 30]. This enables the Bailicai model to progressively learn to differentiate between high- and low-quality retrieval documents during the fine-tuning phase, thereby selectively integrating background information in the process of knowledge acquisition. Finally, we apply low-rank adaptation (LoRA)^[31] fine-tuning on the Meta-Llama3-8B model^[32] to further enhance model performance. Additionally, we meticulously develop a comprehensive knowledge base for retrieval, drawing from sources such as literature, encyclopedias, textbooks, and guidelines. In comparison with existing medical large language models and RAG approaches, Bailicai has exhibited superior performance across multiple medical

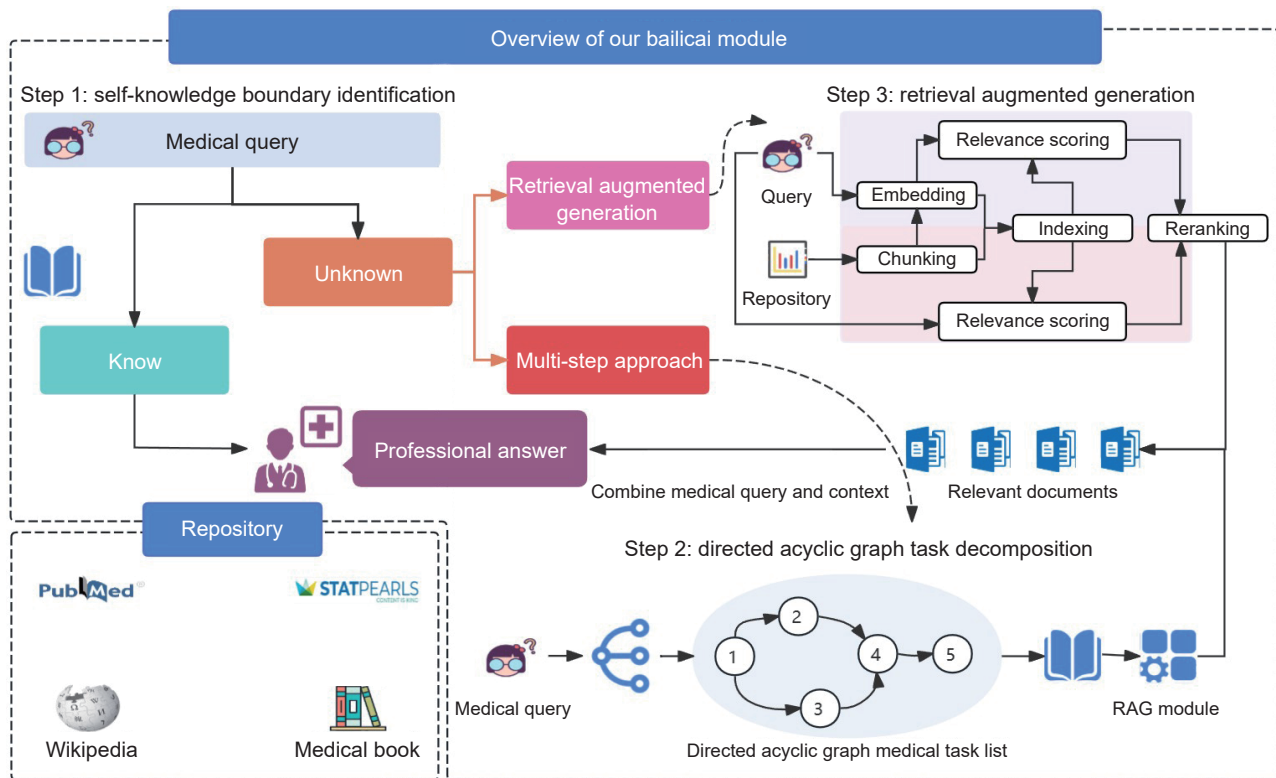


Fig. 1 Bailicai overall process overview.

benchmarks.

Our principal contributions are as follows:

- To the best of our knowledge, this research presents the first implementation of a directed acyclic graph task decomposition paradigm within the retrieval-augmented generation framework, with specific application to the medical domain. This approach utilizes a hierarchical structure to systematically process subtasks, thus providing a more comprehensive and nuanced knowledge base for subsequent RAG applications.

- Leveraging the UltraMedical dataset^[33], we innovatively construct three datasets designed to address subtasks: self-knowledge boundary identification, directed acyclic graph task decomposition, and medical knowledge injection. This construction process employ rigorous data filtering techniques and is innovatively structured around three critical dimensions: data quality, coverage, and necessity, which ensures the overall training effect even when training resources are limited.

- The Bailicai framework integrates the strengths of LLMs and retrieval-augmented generation by coordinating different sub-modules to address the limitations of LLMs in the medical domain. Through

the concept of “hard negative mining”, the LLM learns discrimination skills during the fine-tuning stage, allowing it to focus on documents closely related to the query while avoiding irrelevant content. A series of experiments validated the efficacy of the Bailicai framework in medical applications and demonstrated the contribution of each module.

2 Related Work

2.1 LLM in medical domains

Since the introduction of GPT-3.5 in 2022^[14], LLMs have exhibited breakthrough performance across diverse natural language processing (NLP) tasks^[3, 6, 32]. These models have been adapted for specific domains, producing promising results; notable examples include OceanGPT^[34] in marine science and LawGPT^[35] in legal studies.

In the medical domain, systems are increasingly integrating high-performance proprietary language models, such as OpenAI’s GPT-4 and Google’s PaLM, via API calls in conjunction with prompt engineering techniques. The efficacy of these applications has been validated in several instances, exemplified by GPT-3.5’s performance on the United States Medical Licensing

Examination (USMLE), where it has either met or approached the 60% passing threshold^[36]. MedPrompt, employing dynamic few-shot learning and self-generated reasoning chains, has enhanced GPT-4's accuracy on the USMLE to 90.2%^[7]. Utilizing comparable methodologies, the Med-PaLM model^[37], based on the PaLM architecture^[6], was the first to achieve a 67% accuracy rate on the USMLE, with its subsequent iteration, Med-PaLM 2^[8], attaining an accuracy of 86.5%, commensurate with expert-level performance. However, the implementation of these proprietary models in medical applications raises concerns, particularly regarding potential privacy breaches^[9].

Concurrently, researchers are investigating the potential of open-source large language models, notably the Llama^[32, 38] and Mistral^[39] series, by transforming medical corpora into question-answer pairs through pretraining and fine-tuning processes to enhance performance in the medical domain. Several models, including Med-Alpaca^[13], BioMistral^[12], PMC-LLaMA^[11], and OpenBioLLM^[10], exemplify this approach. PMC-LLaMA, for example, incorporates an extensive corpus of medical papers and textbooks during its pretraining phase to encode domain-specific knowledge, followed by utilizing ChatGPT for instruction alignment during fine-tuning on a range of medical tasks. Notwithstanding these advancements, a significant performance disparity persists between open-source and proprietary models, attributable to the base model's capabilities, training methodologies, and dataset quality. Empirical evidence indicates that high-quality datasets yield more substantial performance improvements compared to larger volumes of lower-quality data^[40]. Furthermore, computational resource constraints have necessitated the adoption of techniques such as low-rank approximation (LoRA)^[31] to mitigate hardware demands during pretraining and fine-tuning phases.

2.2 Retrieval-augmented generation

RAG enhances the performance of large language models by leveraging external knowledge bases to retrieve relevant document segments through semantic similarity computations^[17, 18]. This approach significantly reduces the incidence of hallucinations—defined as instances where generated content deviates from factual accuracy^[15, 16]. Early research on RAG primarily focused on developing

sparse or dense retrievers^[41, 42], whereas contemporary studies have emphasized optimizing the integration of RAG with LLMs. These optimizations encompass the timing of retrieval, methodological enhancements, and refined utilization of contextual information to mitigate noise within retrieved documents^[19, 20].

With respect to adaptive retrieval strategies, the FLARE project has proposed two novel methods: proactive retrieval based on retrieval instructions and confidence-based proactive retrieval, both designed to mitigate unnecessary retrievals^[21]. Furthermore, process optimization has evolved from the “rewrite-retrieve-read” paradigm^[43] to “ITER-RETGEN” implementing iterative retrieval to incrementally access more granular and comprehensive knowledge^[22]. Concerning the management of noise within retrieved documents, studies conducted demonstrated that irrelevant noise documents do not necessarily deteriorate system performance; conversely, they can enhance accuracy by up to 35%. Conversely, documents incorrectly classified as relevant to the query introduce significant interference, substantially impacting the model's generative performance^[44].

In the medical domain, the application of RAG technology remains in its nascent stages. The MEDRAG system assessed performance variations across diverse retrievers and corpora in medical question-answering tasks^[26]. The Self-BioRAG^[45] project integrated Self-RAG^[23] technology in medicine, optimizing it through reflective tokens that address retrieval timing, evaluate the relevance and supporting capacity of retrieved documents in answer generation, and assess the quality of generated outputs. Nevertheless, empirical evidence suggests that this approach fails to surpass the performance of models specifically optimized for medical datasets, potentially attributable to the limitations of smaller models in multi-task integration, as demonstrated by the AUTOACT study^[46]. Furthermore, during the fine-tuning phase, concurrent with data related to RAG tasks, it is imperative to augment the integration of medical knowledge to ensure the model's accurate recognition and utilization of this domain-specific information^[47].

3 Method

In the contemporary medical domain, the integration of RAG technology with large language models remains

in its nascent stage. A significant proportion of research in this area remains exploratory. Identifying the optimal juncture for model-initiated retrieval through adaptive mechanisms presents challenges, while data noise in retrieved information can precipitate erroneous outputs during the generation process. Consequently, leveraging the extensive research of our predecessors, we propose Bailicai, a retrieval-augmented generation framework that has been rigorously refined and optimized at each stage of its development.

3.1 Overview

As illustrated in Fig. 1, our model architecture comprises three fine-tuned large models and one module: $\mathcal{M}_{\text{Medical}}$, $\mathcal{M}_{\text{Know}}$, $\mathcal{M}_{\text{DAG}}$, and $\mathcal{M}_{\text{RAG}}$. Each component executes its designated function within the Bailicai framework, facilitating medical knowledge infusion, self-knowledge boundary identification, directed acyclic graph task decomposition, and retrieval-augmented generation. The framework's development process involves constructing task-specific datasets and optimizing the models for each respective task. During the training phase, synthetic noise was incorporated into the infusion of medical knowledge to augment the model's robustness in leveraging retrieval knowledge. In the ensuing reasoning phase, the process initiates with a user inputting a base query (Q_{Base}). $\mathcal{M}_{\text{Know}}$ initially evaluates whether the query can be addressed using the internal medical knowledge of $\mathcal{M}_{\text{Medical}}$, thereby determining the necessity for external retrieval. If addressable, the answer $\mathcal{A}_{\text{Fin}}$ is generated directly. If not, the query is processed sequentially through $\mathcal{M}_{\text{DAG}}$ and $\mathcal{M}_{\text{RAG}}$ to decompose the question, retrieve, and organize the relevant information. Subsequently, the retrieved information $\mathcal{K}_{\text{RAG}}$, in conjunction with the answer paragraph $\mathcal{K}_{\text{Dirt}}$ (enriched by $\mathcal{M}_{\text{Medical}}$) and the sub-question and its answer $\mathcal{K}_{\text{DAG}}$, are integrated to form comprehensive background information. This integrated composite serves as a new prompt input for $\mathcal{M}_{\text{Medical}}$ to generate the final answer $\mathcal{A}_{\text{Fin}}$.

3.2 Self-knowledge boundary identification

Existing RAG techniques predominantly enhance the input of large language models by incorporating relevant textual segments, aiming to mitigate factual errors in knowledge-intensive tasks. However, this approach exhibits excessive reliance on the efficacy of

the retrieval module and fails to fully leverage the inherent parameterized knowledge of the large language model, potentially compromising output quality. Furthermore, while these techniques may augment task performance, they concurrently introduce significant processing latency, potentially limiting their applicability in real-time scenarios.

Within the Bailicai framework, the $\mathcal{M}_{\text{Know}}$ model conducts an initial assessment to determine if the user's presented medical query can be resolved utilizing the internal parameterized knowledge base of the foundational model, $\mathcal{M}_{\text{Medical}}$. This method identifies the scope, timing, and depth needed for retrieval by determining which aspects of the query are covered by existing knowledge and which require additional information or further decomposition, thus reducing time overhead.

If the user's query Q_{Base} is inferred as “know” by the $\mathcal{M}_{\text{Know}}$ model, it is forwarded to the $\mathcal{M}_{\text{Medical}}$ model, which generates the final response, $\mathcal{A}_{\text{Base}}$. If the result is “unknown”, the query proceeds to the second stage: directed acyclic graph task decomposition. The formalized representation is delineated as follows:

$$\mathcal{M}_{\text{Know}}(Q_{\text{Base}}) = \arg \max_a P(a | \mathcal{P}_{\text{Know}}(Q_{\text{Base}}), \mathcal{M}_{\text{Know}}) \quad (1)$$

where $\mathcal{P}_{\text{Know}}$ indicates the prompt word for the self-knowledge boundary identification stage.

3.3 Directed acyclic graph task decomposition

In the medical domain, addressing complex clinical diagnostic tasks requires meticulous planning of diagnostic protocols, systematic assessment of symptoms against established diagnostic criteria, and implementation of appropriate management strategies. Extant research on task decomposition predominantly employs prompt engineering methodologies (e.g., chains of thought^[48], CoT) to partition tasks into multiple fine-grained subtasks. Through the resolution of these subtasks, requisite information is provided to the primary task, thereby augmenting the quality of the final outcome. However, this task decomposition model may face limitations in effectively articulating and managing complex dependencies inherent in medical tasks. Inspired by research on data science planning^[49], we implement a hierarchical structure of directed acyclic graphs to enhance comprehension and management of the complexity inherent in medical tasks. This structure precisely represents inter-task dependencies, enabling systematic and efficient task

processing. The graph also includes metadata for each task node, such as task description, status, dependencies, and outcomes. The hierarchical directed acyclic graph clarifies task relationships and enhances the efficient management of complex medical tasks.

Specifically, we convey the fundamental question Q_{Base} to M_{DAG} by initializing $\mathcal{P}_{\text{DAG}}$ combinations, subsequently generating a directed acyclic graph question list $Q_{\text{DAG}} = \{Q_1, Q_2, \dots, Q_n\}$. Detailed formulas are provided below to elucidate the methodology,

$$Q_{\text{DAG}} = \arg \max_a P(a | \mathcal{P}_{\text{DAG}}(Q_{\text{Base}}), M_{\text{DAG}}) \quad (2)$$

where $\mathcal{P}_{\text{DAG}}$ indicates the prompt word for the directed acyclic graph task decomposition stage.

Subsequently, we reintegrate the sub-questions into the model M_{Know} to dynamically evaluate the necessity of invoking the external retrieval knowledge module M_{RAG} for knowledge augmentation.

When $M_{\text{Know}}(Q_i) = \text{know}$, the model can address the current question utilizing its intrinsic medical knowledge. We define $\mathcal{A}_i$ as follows:

$$\mathcal{A}_i = M_{\text{Medical}}(Q_i) \quad (3)$$

When $M_{\text{Know}}(Q_i) = \text{unknown}$, meaning the model needs to obtain external knowledge to answer, we define $\mathcal{A}_i$ as follows:

$$\mathcal{A}_i = M_{\text{Medical}}(M_{\text{RAG}}(Q_i) \oplus \hat{A} \oplus Q_i) \quad (4)$$

Among them, the answer $\hat{A}$ corresponds to the set of previous tasks upon which question Q_i depends.

Refer to Section 4.1, for an in-depth discussion on data construction methodologies.

3.4 Medical knowledge injection

Open-source large language models demonstrate the potential to attain expert-level performance in specialized medical tasks through supervised fine-tuning on domain-specific datasets. Conversely, research on integrating RAG with large language models for medical applications remains in its nascent stages. Our findings indicate that the performance enhancement resulting from this integration is contingent upon the intrinsic capabilities of the large model and the domain-specific requirements. Inspired by previous research^[47], we deliberately incorporate golden documents ($\mathcal{D}_*$) and interference documents ($\mathcal{D}_-$) during training to augment the model's capacity to accurately cite golden documents and disregard interference documents in tasks involving medical

literature. We adopt the concept of “hard negative mining”. By vectorizing the documents and calculating the similarity between the questions and the documents, we select those documents with similarity scores between 30 and 100 as interference items ($\mathcal{D}_-$). We classify medical tasks into three categories: medical examination questions, literature-based question answering, and open-ended queries, to comprehensively evaluate the efficacy of RAG in these domains. The methodological details are delineated as follows:

$$\mathcal{D}_{\text{Medical}} : Q_{\text{Base}} + \mathcal{D}_* + \mathcal{D}_- \rightarrow \mathcal{A}_{\text{Fin}} \quad (5)$$

During the ensuing inference phase, the conceptual framework can be expressed as follows:

$$\mathcal{A}_{\text{Fin}} = \arg \max_a P(a | Q, M_{\text{Medical}}) \quad (6)$$

3.5 Retrieval-augmented generation

In the domains of biomedicine and clinical practice, researchers and clinicians frequently leverage supplementary information to address complex problems. Analogously, language models augment their problem-solving capabilities through the retrieval of pertinent documents. This study utilizes the MedCPT model for retrieval. During the construction phase, we implement a dense encoder, $E_P(\cdot)$, which maps text paragraphs to d -dimensional real vectors and generates an index database comprising M paragraphs for retrieval. In the query process, another encoder, $E_Q(\cdot)$, transforms the input question into a d -dimensional vector, which is subsequently utilized to retrieve the K paragraphs exhibiting maximum similarity to the question vector. The similarity between the question vector and document vectors is computed via the dot product. The detailed implementation methodology is delineated as follows:

$$\text{sim}(Q_{\text{Base}}, \mathcal{P}_{\text{RAG}}) = E_Q(Q_{\text{Base}})^T E_P(\mathcal{P}_{\text{RAG}}) \quad (7)$$

4 Experimental

4.1 Bailicai medical dataset

The bailicai medical dataset comprises both a training dataset and a retrieval dataset. The training dataset, extracted from the UltraMedical Dataset, encompasses approximately 277 677 high-quality entries of medical instruction data. UltraMedical incorporates over 410 000 meticulously curated medical instructions, integrating

both manually crafted and algorithmically-generated synthetic prompts. The construction of UltraMedical involves the extraction of high-quality data from medical examination databases and PubMed articles, supplemented by synthetic prompts generated through sophisticated algorithms, including self-evolution and heuristic-based filtering^[33]. The medical instruction data illustrated in Fig. 2 is elaborated upon:

Recent studies have elucidated that during the instruction execution phase, a limited quantity of high-quality data typically yields superior performance compared to a large volume of low-quality data. We commence our analysis with the UltraMedical dataset, utilizing the model's score on the original data to establish a threshold of nine points for initial screening. we employ the model-oriented instruction data selection (MODS) method to further filter data from the perspectives of both data quality and coverage (e.g., similarity), thereby ensuring that training costs and performance are balanced through data engineering

when server resources are limited. Specifically, the OpenAssistant reward-model-debertav3-large-v2 model (a reward model developed using the DeBERTa architecture, which stands for decoding-enhanced BERT) is utilized to evaluate data quality and filter data by applying a threshold value of 1. In addition, to prevent the selection of highly similar quality data, the k-center-greedy algorithm is employed to select a subset of data points that maximizes the inter-point distance.

Subsequently, using the seed dataset, data is derived and filtered according to three distinct sub-tasks, facilitated by self-instruct. Specifically:

(1) Self-knowledge boundary identification A random subset of questions is extracted from the baseline dataset, and the Meta-Llama-3-70B model is utilized to perform task-driven adaptation on this subset, focusing on a specialized medical instruction dataset. Initially, we acquire and curate relevant data in accordance with predefined prompts (see Fig. 3) for

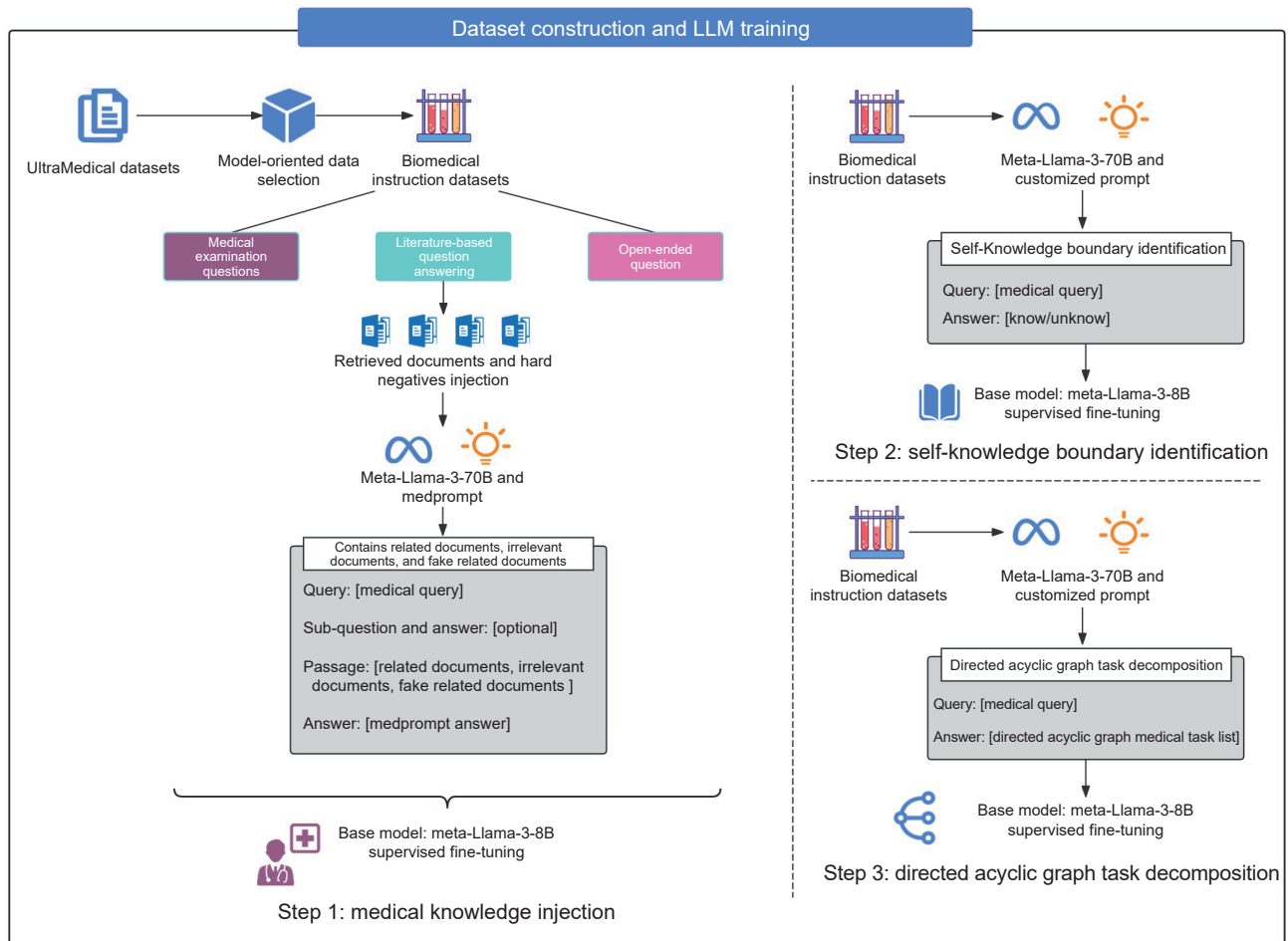


Fig. 2 Diagram of the bailcai dataset construction and training process.

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=====Prompt=====
Below is an instruction that describes a task. Write a response that appropriately completes the request.

### Instruction:
Based on your inner knowledge, determine whether you can answer this medical question. If you are sure you can answer it, please use "know" as your only answer. If you cannot answer it or are not confident about it, please use "unknow" as your only answer. If the actual situation is that you cannot answer it or the answer is incorrect, but you say "know", you will be severely punished.

Medical question: {query}

### Response:

```

Fig. 3 Prompt template for self-knowledge boundary identification with Bailicai.

preliminary knowledge distillation. Subsequently, a rigorous data cleaning process is implemented to eliminate inconsistencies and redundant information from the dataset. This process includes aggregating data based on the predefined prompts, followed by manual sampling scores obtained using GPT-4 to ensure quality. Upon completion, the resulting dataset comprise 71 557 data points.

(2) Directed acyclic graph task decomposition

Similarly, we adhered to the preset improvement protocol (see Fig. 4). Following this series of steps, the resulting dataset comprised 32 849 data points.

(3) Medical knowledge injection By incorporating distracting documents during the training process, the model can more effectively focus on documents relevant to the question while ignoring irrelevant information. When interference documents are included in the original dataset, the MedPrompt strategy is applied to enhance the model's reasoning capabilities using dynamic few-shot learning, self-generated chain of thought, and choice shuffling ensemble methods, while documenting its responses. The resulting dataset after these refinements comprised 173 271 data points.

(4) Retrieval-augmented generation Raw data has been collected from four distinct sources to enhance retrieval generation. The sources include PubMed (biomedical abstracts), Wikipedia (online encyclopedia), StatPearls (medical education database), and medical textbooks. Furthermore, MedCPT is employed to train an integrated retrieval and re-ranking model using 255 million query records from PubMed search logs, utilizing a comparative training method. Text paragraphs are transformed into vectors and stored in the Faiss vector database, facilitating efficient

subsequent retrieval. Table 1 provides comprehensive statistics regarding the medical retrieval corpus and the number of indexed documents.

4.2 Training detail

This study employs the Bailicai dataset, which is processed to conform to the Alpaca dataset format for fine-tuning on various tasks. Supervised fine-tuning (SFT) was conducted on the Meta-Llama-3-8B model using four NVIDIA Tesla V100 graphics processing units (GPUs), implementing the LoRA method. LoRA utilizes a low-rank decomposition matrix to approximate parameter updates for each network layer, potentially minimizing the number of parameters required for training when adapting to downstream tasks^[31]. The cross-entropy loss function is employed to calculate the loss value during the fine-tuning process. The maximum context lengths for models M_{Know} , M_{DAG} , and M_{Medical} are configured to 1024, 2048, and 2816 tokens, respectively. All three models employ the AdamW optimizer and undergo domain-specific fine-tuning in the medical field for three epochs, with a batch size of 2 and a learning rate of 2×10^{-5} . Furthermore, the LoRA hyperparameters—rank (r), alpha (α), and dropout—are set to 128, 256, and 0.1, respectively. Here, r denotes the rank of the low-rank matrices, α represents a scaling factor for the low-rank update, and dropout refers to the rate at which units are randomly dropped during training to help mitigate overfitting.

4.3 Retrieval-augmented generation

The RAG module employs a two-stage retrieval strategy, consisting of coarse-grained and fine-grained retrieval phases.

The coarse-grained retrieval stage initiates with

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=====Prompt=====

Below is an instruction that describes a task. Write a response that appropriately completes the request.

### Instruction:
Please analyze the medical question provided and try to break the question into smaller, distinct sub-questions. By subsequently solving the sub-problems and combining the answers to the sub-problems, the original question can be better answered. A problem consists of one to four sub-problems.

When posing questions based on patient symptom data, all sub-questions should be expanded from the given options to include corresponding background information, rather than directly asking questions based on the patient's symptom data. If the problem can be decomposed, carefully follow the instruction, don't make unnecessary changes and do not answer any sub-questions.

Output a list of Json following the format:

[
{
  "task_id": "unique identifier for a sub-problem, can be an ordinal",
  "dependent_task_ids": ["The sub-issue ID of the prerequisite for this sub-issue"],
  "instruction": "what you should do in this sub-problem, one short phrase or sentence"}, ...
]

If the problem does not need to be broken down into sub-problems, your output format is as follows:

[
{
  "task_id": "1",
  "dependent_task_ids": [],
  "instruction": original question
}
]

Medical question: {query}

### Response:

```

Fig. 4 Prompt template for directed acyclic graph task decomposition with Bailicai.

Table 1 Structural and statistical features of the indexed medical corpus.

Data	Number of chunks	Embedding	Introduction
PubMed	23.9M	80.1G	Medical literature
Wikipedia	29.9M	100.3G	General knowledge
StatPearls	301.2k	900M	Medical database
Textbooks	125.8k	368.7M	Medical education
Merge	54.2M	181.7G	Knowledge combination

Note: "Embedding" denotes the file size resulting from processing with the MedCPT model. k denotes 10^3 , M denotes 10^6 , and G denotes 10^9 .

information extraction from four medically-related corpora. MedCPT, a unified retrieval tool, vectorizes all texts in the corpus. To optimize retrieval efficiency for large-scale data, a hierarchical navigable small world (HNSW) index is implemented within the Faiss vector database. The retrieval process commences at the top layer, utilizing a greedy algorithm to identify the node closest to the query vector. Subsequently, the

search progresses downward, exploring neighboring nodes at each layer until reaching the bottom layer.

The fine-grained retrieval stage employs a reranking model (bge-reranker-large) to score the top 32 retrieved paragraphs. The scoring criteria encompass semantic consistency, answer completeness, logical coherence, and relevance to the user query. Ultimately, the five highest-scoring documents are selected for further

processing.

4.4 Medical domain benchmark

To comprehensively evaluate the domain expertise of the Bailicai-based large language model in the medical field and its retrieval augmentation efficacy, this study utilizes five medical question-answering benchmark datasets^[26, 37]: MedQA, MedMCQA, MMLU-Med, PubMedQA, and BioASQ. The study evaluates the model's overall performance in medical question-answering for each dataset and task by computing its predictive accuracy. Detailed characteristics of the specific datasets are presented in Table 2.

4.5 Baseline

To comprehensively assess and compare the Bailicai framework, benchmarks were conducted utilizing a standardized medical evaluation protocol, retrieval model, and a domain-specific medical retrieval corpus. The study encompasses baseline models and methods from three domains: general-purpose, domain-specific, and RAG.

(1) Chain-of-thought^[48] This method facilitates step-by-step reasoning through prompts such as “Let’s think step by step”, thereby enhancing problem-solving accuracy.

(2) Med-alpaca^[13] This model is fine-tuned on Llama using a curated dataset of over 160 000 medical domain-specific entries, comprising medical literature and authentic question-and-answer (Q&A) pairs.

(3) BioMistral^[12] This model, based on the mistral architecture, undergoes additional pre-training on an extensive corpus of medical literature from PubMed central to enhance its proficiency in medical terminology and question-handling. It is subsequently

fine-tuned on the MedQA, PubMedQA, and MedMCQA datasets.

(4) PMC-LLaMA^[11] This model incorporates a comprehensive corpus comprising 4.8 million biomedical academic papers and 30 000 medical textbooks, enhancing its medical knowledge integration. Furthermore, it undergoes domain-specific instructional tuning on a 202-million-token dataset, encompassing medical Q&A and knowledge graphs.

(5) OpenBioLLM^[10] Fine-tuned on Meta-Llama-3-8B using a comprehensive medical instruction dataset, this model incorporates direct preference optimization during fine-tuning, achieving high performance in multi-task scenarios including clinical note summarization and medical Q&A.

(6) Flan-PaLM^[37] This variant of the 540-billion parameter PaLM model is fine-tuned on MultiMedQA, utilizing strategies such as few-shot prompts, Chain-of-thought, and consistency prompts. It demonstrates state-of-the-art performance across multiple medical Q&A datasets.

(7) Med-PaLM2^[8] This model integrates an enhanced foundational LLMs (PaLM 2), domain-specific fine-tuning, and a novel ensemble refinement prompting strategy. It effectively bridges the gap with clinical responses, demonstrating significant improvements across various medical benchmarks.

(8) Mistral-7B-V0.3^[39] MistralAI’s third-generation model expands the vocabulary and introduces new functionalities, such as function calls, compared to its predecessors.

(9) Meta-Llama-3^[32] Meta’s third-generation model significantly expands its vocabulary and implements grouped query attention (GQA). It employs diverse filtering technologies, including heuristic

Table 2 Descriptive statistics of medical domain benchmarks.

Dataset	Format	Option	Total number	Concise description	Source
MedQA	Question + answer	a/b/c/d	1273	USA medical licensing examination	Examination
MedMCQA	Question + answer + explanations	a/b/c/d	4183	Medical entrance examinations in India	Examination
MMLU-Med	Question + answer	a/b/c/d	1089	Encompasses biomedical knowledge, including anatomy, clinical knowledge, and professional medicine, etc.	Examination
PubMedQA	Question + context + answer	Yes/no/maybe	500	Biomedical literature question answering	Ref. [50]
BioASQ	Question + answer	Yes/no	618	Biomedical semantic indexing and question answering systems encompass a wide range of medical subfields	Ref. [51]

Note: a, b, c, and d represent the options of the multiple choice question.

filters, Not suitable for work (NSFW) filters, and semantic deduplication, to enhance training data quality. The model also incorporates advanced techniques such as supervised fine-tuning (SFT), rejection sampling, proximal policy optimization (PPO), and direct preference optimization (DPO).

(10) **ChatGPT**^[3] Launched by OpenAI in November 2022, this commercial model demonstrates superior performance across diverse domains and natural language processing tasks.

(11) **Self-BioRAG**^[45] This model adapts the Self-RAG framework to the medical domain, utilizing GPT-4 to generate specific reflection markers for knowledge distillation. It manages various aspects of the retrieval process, including retrieval decisions and content relevance assessment.

(12) **MEDRAG**^[26] This approach integrates multiple domain-specific medical corpora and conducts comprehensive evaluations across various retrieval models and large language models.

5 Result and Discussion

This study presents a series of experiments designed to assess the performance of the Bailicai model on

medical domain tasks in comparison with baseline models. The evaluation encompass comparisons between Bailicai and models specifically fine-tuned for the medical domain, as well as models and frameworks optimized for retrieval-augmented generation tasks. Experimental findings indicate that Bailicai exhibits substantial advantages in processing medical-related tasks. Furthermore, a series of ablation studies are conducted to quantify the specific impact of various components on the model's overall performance.

5.1 Main result

Comparison with large models in the medical domain

Table 3 demonstrates the superior performance of the Bailicai model across five medical benchmarks. In comparison with domain-specific models of smaller scale (e.g., 7B, 8B, and 13B parameters, where “B” stands for billion) that are pre-trained and fine-tuned for medical tasks, Bailicai attains state-of-the-art results on benchmarks including MedQA, MMLU-Med, PubMedQA, and BioASQ. On the MedMCQA benchmark, Bailicai's performance is comparable to OpenBioLLM, with a marginal difference of 0.14% and an average score of 71.82%.

Table 3 Evaluation of accuracy and overall results on five tasks.

Model		Number of parameters	Accuracy (%)					
			MedQA	MedMCQA	MMUL-Med	PubMedQA	BioASQ	Average
Medical large language model (CoT)	MedAlpaca	7B	37.63	32.70	57.58	49.80	62.62	48.07
	BioMistral	7B	43.21	41.33	59.32	42.00	70.23	51.24
	PMC-LLaMA	13B	56.40	47.55	51.88	61.60	66.34	56.75
	OpenBioLLM	8B	57.11	56.03	75.11	71.80	59.87	63.98
	Flan-PaLM [‡]	540B	67.60	57.60	79.16	79.00	–	70.84
	Med-PaLM2 [‡]	340B	83.70	71.50	88.43	74.00	–	79.41
Universal large language model (CoT)	Mistral-7B-v0.3	7B	48.78	45.35	63.64	48.00	68.93	54.94
	Meta-Llama-3	8B	62.69	52.52	72.73	75.40	72.01	67.07
	Meta-Llama-3	70B	76.98	65.77	84.30	76.80	79.13	76.60
	ChatGPT-3.5 [‡]	–	57.71	53.79	70.84	72.66	74.27	65.85
	ChatGPT-4 [‡]	–	78.80	69.50	87.20	70.00	84.30	79.00
Retrieval-augmented generation	Meta-Llama-3-RAG	8B	57.97	55.27	70.71	76.40	81.55	68.38
	Self-BioRAG	13B	48.60	44.00	57.20	54.60	–	51.10
	ChatGPT-3.5-MEDRAG [‡]	–	66.61	58.04	75.48	67.40	90.29	71.56
	ChatGPT-4-MEDRAG [‡]	–	82.80	66.65	87.24	70.60	92.56	79.97
	Bailicai	Our model	8B	66.85	55.89	76.12	77.40	82.85

Note: Each baseline model is evaluated under standardized settings. The experimental protocol consistently utilized a chain-of-thought strategy in prompts, with the temperature parameter maintained at 0. To retrieval-augmented generation, both the MedCPT model and the PubMed knowledge base are utilized. B stands for billion. Bold values denote the superior performance among all methods for each model. [‡] signify results reported in the original paper.

Moreover, Bailicai (8B) demonstrates performance parity with larger models such as Flan-PaLM (540B) in the medical domain, a characteristic of particular importance in resource-constrained healthcare environments due to its capability for local deployment coupled with the absence of privacy leakage concerns. These findings underscore Bailicai's substantial advantages in processing diverse medical tasks.

Comparison with general domain large models

As evidenced in Table 3, Bailicai outperforms ChatGPT-3.5 (achieving an average score of 65.85%) by 5.97%. In the medical domain, given the critical nature of user privacy, Bailicai offers a significant advantage through its capability for local deployment. Furthermore, Bailicai achieves superior performance relative to the Meta-Llama-3-70B model on the PubMedQA and BioASQ benchmarks, surpassing it by 0.6% and 3.72% respectively.

Comparison with retrieval-augmented generation models and methods Tables 3 and 4 illustrate that Bailicai effectively mitigates hallucination issues stemming from noise in retrieved documents through retrieval-enhanced generation, outperforming other retrieval-augmented generation models. Notably, in comparison to the Self-BioRAG model, which is specifically fine-tuned for retrieval-augmented generation, Bailicai demonstrates a significant advantage across multiple datasets, yielding an average performance improvement of 20.72%.

5.2 Ablation study

To evaluate the efficacy of various components within the Bailicai framework, a series of ablation studies are conducted, as detailed in Table 4. Each experiment incorporates the integration of distinct modules:

(1) Medical knowledge injection This variant implements domain-specific fine-tuning on the Bailicai medical dataset, encompassing medical examinations, literature-based research, and open-domain medical question answering.

(2) Self-knowledge boundary identification This module determines whether a query can be resolved solely using the medical knowledge parameterized within the model.

(3) Directed acyclic graph task decomposition This variant enhances the model's capacity to process complex queries by decomposing them into a sequence of structured sub-tasks.

(4) Retrieval-augmented generation This variant incorporates RAG technology to manage unknown or challenging queries. The process entails retrieving relevant documents and embedding them alongside the query in the prompt, thereby augmenting external knowledge.

Table 4 delineates the results of ablation studies, revealing that the removal of any module from the Bailicai framework leads to a performance decline, thus highlighting the critical contributions of each module to the overall efficacy. In comparison to the baseline model, Meta-Llama-3-8B, which utilizes Retrieval-Augmented Generation technology, the Bailicai framework demonstrates superior performance in information extraction and increased robustness against distracting documents. Concurrent application of all four modules yields performance improvements of 8.88% and 5.41% on the MedQA and MMLU-Med benchmarks, respectively, with an average score increase of 3.44%.

Experimental results indicate that the integration of RAG technology with the directed acyclic graph task

Table 4 Ablation study and accuracy evaluation of Bailicai component efficacy.

Component module		Accuracy (%)					
		MedQA	MedMCQA	MMUL-Med	PubMedQA	BioASQ	Average
Meta-Llama-3-8B	Basic model	62.69	52.52	72.73	75.40	72.01	67.07
	Basic model + RAG	57.97	55.27	70.71	76.40	81.55	68.38
Bailicai	Medical	65.67	55.82	75.85	77.40	78.80	70.71
	Medical + boundary + RAG	66.06	56.08	75.67	78.40	81.72	71.59
	Medical + boundary + DAG	65.75	55.85	75.85	77.20	80.23	70.98
	Medical + boundary DAG + RAG	66.85	55.89	76.12	77.40	82.85	71.82

Note: To enhance retrieval performance, both the MedCPT model and the PubMed knowledge base are utilized. In the table, "medical" refers to "medical knowledge injection", "boundary" represents "self-knowledge boundary", "DAG" stands for "directed acyclic graph task decomposition", and "RAG" denotes "retrieval-augmented generation." Bold values denote the superior performance among all methods for each model.

decomposition module optimizes overall performance, achieving 71.82%. This combined approach results in a marginal 0.23% increase in overall performance compared to the isolated use of RAG technology, while concurrently producing a 1.0% performance decline on the PubMedQA benchmark. This performance decrease may be attributed to the inherent structure of the PubMedQA dataset, which incorporates key contextual documents essential for question answering. Document retrieval following task decomposition potentially generates token lengths exceeding the maximum threshold (2812) established during the medical knowledge injection phase of Bailicai, consequently impacting performance negatively.

To test this hypothesis, the experiments on the PubMedQA benchmark are reconfigured, excluding the golden context documents provided in the original dataset and focusing exclusively on question-based evaluation. As evidenced in Table 5, the incremental addition of modules yields an improvement in Bailicai's performance on the PubMedQA benchmark from 52.60% to 69.80%, corroborating the hypothesized causes of the initial performance decline.

The integration of the directed acyclic graph task decomposition module with RAG technology results in performance improvements of 0.86% and 1.3% on MedQA and BioASQ tasks, respectively, relative to the isolated application of RAG. These performance enhancements are predominantly attributed to the

Table 5 Comparative analysis of Bailicai component performance on the PubMedQA[®].

Module	PubMedQA (%)
+ Medical knowledge injection	52.60
+ DAG task decomposition	53.80
+ Retrieval-augmented generation	69.80

Note: PubMedQA[®] excludes original golden documents, retaining only the query and answer.

structured decomposition of complex queries, facilitating more effective access to a broader spectrum of background information by the retrieval-augmented generation technology. This observation suggests that decomposing queries into hierarchical sets of sub-tasks optimizes the retrieval process, consequently improving the relevance and accuracy of the retrieved information.

Consequently, the synergistic interaction of these four modules not only improves overall performance on medical domain tasks but also effectively attenuates the adverse effects of irrelevant retrieval content, thus enhancing the efficacy of RAG technology in medical applications.

5.3 Comparative corpus retrieval

The efficacy of RAG systems being intrinsically linked to the quality of the retrieval corpus, this study investigates the influence of diverse retrieval corpora on the Bailicai model's performance across multiple medical benchmarks. Employing the Bailicai model, exclusively fine-tuned with medical knowledge injection as a baseline, the study evaluated the effects of five distinct corpora. As evidenced in Table 6, the PubMed corpus demonstrates the highest average score of 71.58% among the five corpora, with Wikipedia achieving a score of 70.87%. These results align with anticipated outcomes, underscoring the inherent content limitations of each corpus. Specifically, PubMed, a platform primarily utilized for retrieving biomedical and life science literature, provides comprehensive and current content, encompassing cutting-edge and in-depth medical knowledge. Conversely, Wikipedia, an online encyclopedia covering a broad spectrum of topics from science to history and popular culture, offers substantial information volume but demonstrates limited depth in the medical domain. Relative to the baseline, the PubMed corpus exclusively exhibited performance

Table 6 Comparative analysis of the performance of Bailicai in different medical retrieval corpora.

Medical retrieval corpora		Accuracy (%)					
		MedQA	MedMCQA	MMLU-Med	PubMedQA	BioASQ	Average
+ Medical knowledge injection		65.67	55.82	75.85	77.40	78.80	70.71
+ Retrieval-augmented generation	PubMed	66.06	56.08	75.67	78.40	81.72	71.58
	Wikipedia	65.91	55.75	76.40	77.00	79.29	70.87
	StatPearls	64.73	55.39	76.03	75.60	79.13	70.18
	Textbooks	65.83	55.70	75.30	77.40	77.99	70.44
	Merge	64.26	56.08	76.03	79.00	81.88	71.45

Note: "Medical knowledge injection" is used as a contrast to "retrieval-augmented generation" when RAG is not used.

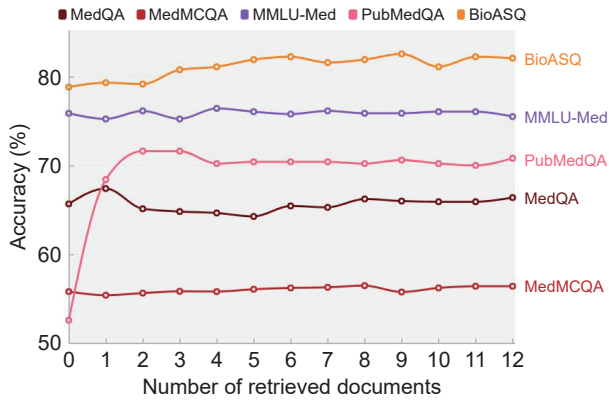


Fig. 5 Performance of Bailicai on various benchmarks with different retrieved documents.

improvements across four datasets, with a marginal decline of 0.18% on the MMLU-Med dataset, thus substantiating the robustness of its corpus.

5.4 Comparative document retrieval volume

The efficacy of RAG systems is substantially modulated by the quantity of documents retrieved. Figure 5 elucidates a pronounced curve for the Bailicai system, delineating the correlation between the number of retrieved medical documents and accuracy across diverse medical benchmarks. The graph reveals that Bailicai maintains notable performance even with a minimal number of retrieved documents (e.g., one document), despite limited RAG-derived information, a phenomenon attributable to the medical knowledge integrated during the fine-tuning phase. As the number of retrieved documents increases, Bailicai’s accuracy exhibits marginal enhancement and stabilization, consequent to the continuous augmentation of background information. This trend deviates from typical observations in RAG studies, where accuracy often fluctuates with increasing document retrieval, thus accentuating the robustness of the Bailicai system. Moreover, given the custom token length limit (2812) established during the fine-tuning phase, surpassing a specified threshold in the number of retrieved documents is hypothesized to induce a decline in

accuracy.

5.5 Time expense analysis

In current research on RAG, retrieval is universally applied to all queries, neglecting the parameterized knowledge inherent in the model. While some studies have implemented a “divide and conquer” approach, the processing of sub-queries remains deficient in targeted retrieval, resulting in inefficient resource allocation and increased computational time. To address this limitation, the Bailicai framework utilizes a model for self-knowledge boundary identification to ascertain whether an incoming query can be resolved without external knowledge, thus adaptively invoking RAG, analogous to a valve-switch mechanism. This methodology aims to improve computational efficiency and optimize resource allocation.

Table 7 illustrates the classification statistics for “known” and “unknown” queries, in addition to the average number of sub-queries per benchmark test, as determined by the self-knowledge boundary identification model. By leveraging this model, the Bailicai framework can allocate resources more efficiently to complex and challenging queries, thereby reducing the frequency of retrieval system calls and improving performance when integrated with other frameworks.

Moreover, the MedQA benchmark, which encompasses extensive patient inquiry data, exhibited the highest average number of sub-queries at 3.56. This benchmark illustrates the model’s proficiency in distinguishing and decomposing query complexity. The decomposition of patient inquiries into granular sub-tasks enables the collection of critical clinical indicators necessary for diagnosis, thus improving the efficacy of problem resolution. These results are further substantiated in Table 4.

6 Conclusion

In this paper, we present the Bailicai framework, a

Table 7 Statistics of “know” and “uknow” categories and average number of sub-queries across various medical benchmarks on Bailicai.

Benchmark	Number of “know”	Number of “uknow”	Average number of sub-queries
MedQA	807	466	3.56
MedMCQA	3583	600	2.64
MMLU-Med	956	133	3.20
PubMedQA	169	331	1.78
BioASQ	217	401	2.81

retrieval-augmented generation framework specifically tailored for the medical field to improve medical outcomes. The Bailicai framework is engineered to autonomously perform retrieval tasks and decompose complex queries into a structured hierarchy of sub-problems. Furthermore, this framework incorporates content evaluation mechanisms to mitigate hallucination phenomena prevalent in large language models utilized in medical applications, while concurrently reducing noise stemming from irrelevant or spurious documents typically encountered in conventional retrieval-augmented generation methodologies. Empirical evaluations across diverse medical benchmarks corroborate the superior performance of the Bailicai framework. Ablation studies further substantiate the efficacy of the framework's constituent components.

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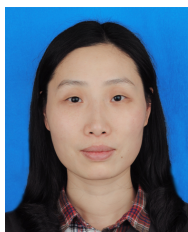
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